

PAPER CODE	EXAMINER	DEPARTMENT	TEL
INT102	Wenjin Lu	Intelligent Science	1505

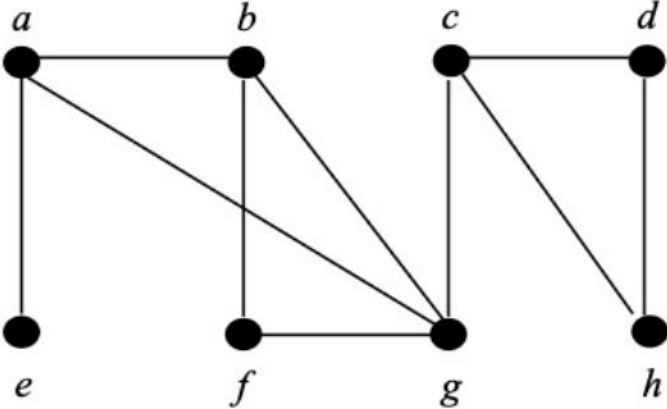
2nd SEMESTER 2021/22 EXAMINATIONS (RESIT Open Book)**BACHELOR DEGREE – Year 2****ALGORITHMIC FOUNDATIONS AND PROBLEM SOLVING****TIME ALLOWED: 2 Hours**

INSTRUCTIONS TO CANDIDATES**READ THE FOLLOWING CAREFULLY:**

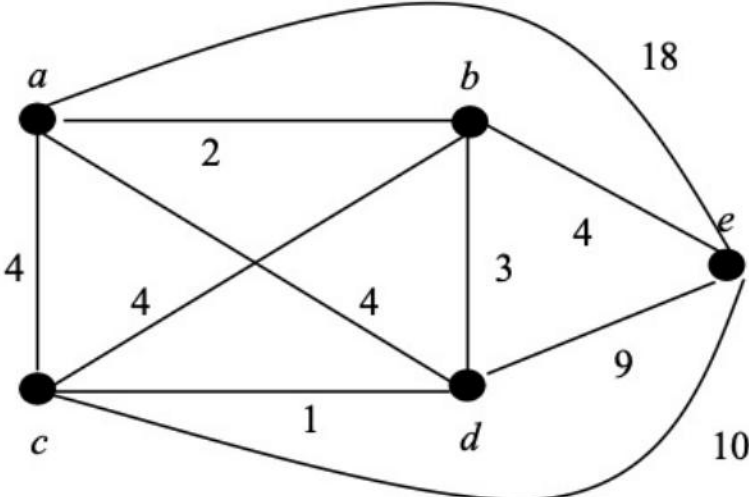
1. The paper consists of Part A and Part B. Answer all questions in both parts. Total marks available are 100. Marks for this examination account for 80% of the total credit for INT102.
2. In Part A, each of the questions comprises 5 statements, for which you should select the one most appropriate answer.
3. Answers to questions in Part B should be written in the answer script.
4. This is an OPEN BOOK examination. You can reference textbooks and notes but discuss with other students in any way is not allowed.
5. The time of the exam is strictly limited to 2 hours.
6. For students who take the exam online, at the end of the examination, be absolutely sure to submit your answer via Learning Mall. The time for submission of your answer via Learning Mall is strictly limited to 15 minutes. Once the time is over, the submission link will be closed.
7. All answers must be in English.

1. Two main measures for the efficiency of an algorithm are	2.5
a) Processor and memory b) Complexity and capacity c) Time and space d) Data and space e) Data and memory	
2. Considering the following algorithm	2.5
<pre> input m count = 0 x = 1 while x < m do begin x = x * 2 count = count + 1 end output count </pre> What is the output of the algorithm for $m=2k+1$? a) m b) $k+1$ c) $k-1$ d) k e) $2k$	
3. The time factor when determining the efficiency of an algorithm is measured by	2.5
a) Counting the microseconds b) Counting the number of key operations c) Counting the number of statements d) Counting the kilobytes of the algorithm e) Counting the number of variables	
4. The space factor when determining the efficiency of an algorithm is measured by	2.5
a) Counting the memory needed by the algorithm b) Counting the number of statements c) Counting the kilobytes of the algorithm d) Counting the maximum disk space needed by the algorithm e) Counting the number of key operations	
5. Running time $T(n)$, where 'n' is input size of a recursive algorithm is given as follows:	2.5

Which of the following algorithms, its time complexity can be expressed by the $T(n)$?	
a) Binary Search b) Merge Sort c) Insertion Sort d) Selection Sort e) Dijkstra's algorithm	
6. Considering the running time $T(n)$ given above in the question 5. Order of magnitude of $T(n)$ is	2.5
a) $O(n^2)$ b) $O(n)$ c) $O(n \log n)$ d) $O(n^3)$ e) $O(n^n)$	
7. The worst case occurs in a linear search algorithm when	2.5
a) Item is somewhere in the middle of the array b) Item is not in the array at all c) Item is the last element in the array d) Item is the last element in the array or is not there at all e) None of the above	
8. The best case occurs in a linear search algorithm when	2.5
a) Item is the first element in the array b) Item is not in the array at all c) Item is the last element in the array d) Item is not in the array at all e) None of the above	
9. The time complexity of a linear search algorithm is	2.5
a) $O(n)$ b) $O(\log n)$ c) $O(n^2)$ d) $O(n \log n)$ e) $O(1)$	
10. Running time $T(n)$, where n is input size of a recursive algorithm is given as follows: $T(n) = \begin{cases} 1 & \text{if } n = 1 \\ T(n/2) + 1 & \text{if } n > 1 \end{cases}$	2.5

The time complexity of the algorithm is a) $O(n^2)$ b) $O(n \log n)$ c) $O(n^3)$ d) $O(n^n)$ e) $O(\log n)$	
Questions 11 to 13 refer to the following graph:	
	
11. The total degree of the graph is a) 21 b) 18 c) 20 d) 19 e) 10	2.5
12. Starting at the vertex <i>a</i> and resolving ties by the vertex alphabetical order, traverse the graph by breadth-first-search (BFS). Then, the order of vertices visited is a) <i>a, b, d, e, f, c, h, g</i> b) <i>a, b, c, d, e, f, g, h</i> c) <i>a, b, f, g, c, d, h, e</i> d) <i>a, b, e, g, f, c, d, h</i> e) None of the above	2.5
13. Starting at the vertex <i>a</i> and resolving ties by the vertex alphabetical order, traverse the graph by depth-first-search (DFS). Then, the order of vertices visited is a) <i>h, g, f, e, d, c, b, a</i>	2.5

b) a, b, f, g, c, d, h, e c) a, b, e, g, f, c, d, h d) a, b, d, c, e, f, g, h e) None of the above	
14. Let T be a tree constructed by Dijkstra's algorithm in the process of solving the single-source shortest path problem for a weighted connected graph G.	2.5
I. T is a spanning tree of G	
II. T is a minimum spanning tree of G	
III. T is a binary tree	
Which one of the following is correct?	
a) I is true, II and III are false	
b) I and II and III are true	
c) I and II and III are false	
d) II and III are true but I is false	
e) None of the above	
15. Let G be a weighted connected graph	2.5
I. If e is a minimum-weight edge in G, it must be contained in a MST.	
II. If e is a minimum-weight edge in G, it must be contained in each MST.	
III. If e is a maximum-weight edge in G, it must not be contained in any MST.	
Which one of the following is correct?	
a) I and III are true, II is false	
b) I and II and III are true	
c) I and II and III are false	
d) II and III are correct but I is false	
e) None of the above	
16. Let G be a weighted connected graph	2.5
I. If the edge weights are all different G must have exactly one MST	
II. If the edge weights are not all different G must have more than one MST	

Which one of the following is correct?	
a) I and III are true, II is false b) I and II and III are true c) I and II and III are false d) II and III are true but I is false e) None of the above	
Questions 17 to 19 refer to the following weighted connected graph graph:	
	
17. Let T be a minimum spanning tree of the graph computed using Kruskal's algorithm. The order of edges selected by Kruskal's algorithm is	2.5
a) (c,d) (a,b) (b,d) (a,c) b) (c,d) (a,b) (b,d) (b,e) c) (c,d) (a,b) (b,d) (a,d) d) (c,d) (a,b) (b,d) (b,c) e) None of the above	
18. Let T be a minimum spanning tree of the graph computed using Prim's algorithm. Assume vertex a is selected first, then the order of vertices selected by Prim's algorithm is	2.5
a) a, b, d, e, d b) a, c, d, b, e c) a, d, c, e, b d) a, b, d, c, e e) None of the above	
19. Assume the source vertex is a . Running Dijkstra's algorithm for the graph, after the termination, the labels for vertices are	2.5
a) $a(0,-)$, $b(2,a)$, $c(4,a)$, $d(5,b)$, $e(6,b)$ b) $a(0,-)$, $b(2,a)$, $c(4,a)$, $d(5,c)$, $e(6,b)$ c) $a(0,-)$, $b(2,a)$, $c(4,a)$, $d(4,a)$, $e(6,b)$	

d) $a(0,-)$, $b(2,a)$, $c(6,b)$, $d(4,a)$, $e(6,b)$ e) None of the above	
20. Assume the source vertex is a. Running Dijkstra's algorithm for the graph, after the termination, which one of the followings could be an order of vertices selected by Dijkstra's algorithm? a) a, b, e, c, d b) a, b, d, e, c c) a, b, d, c, e d) a, b, e, d, c e) None of the above	2.5
Questions 21 to 24 refer to the following Selection sort algorithm.	
ALGORITHM <i>SelectionSort</i>($A[0..n-1]$) //Sorts a given array by selection sort //Input: An array $A[0..n-1]$ of orderable elements //Output: Array $A[0..n-1]$ sorted in ascending order for $i=0$ to $n-2$ do $min = i$ for $j=i+1$ to $n-1$ do if $A[j] < A[min]$ $min = j$ if $i < min$ do swap $A[i]$ and $A[min]$	
21. The time complexity of the Selection sort algorithm is a) $O(n \log n)$ b) $O(2^n)$ c) $O(n^2)$ d) $O(2n)$ e) None of the above	2.5
22. The number of key comparisons needed to sort the numbers $A[0..5] = [6, 5, 4, 3, 2, 1]$ in ascending order using the selection sort algorithm is a) 10 b) 4 c) 3 d) 15 e) 20	2.5
23. The number of swapping operations needed to sort the numbers $A[0..5] = [6, 5, 4, 3, 2, 1]$	2.5

- a) 10
- b) 4
- c) 3
- d) 15
- e) 20

24. To merge the following two sorted sequences into a single sorted sequence, using the Merge algorithm given in the lecture,

2.5

(3, 8, 10, 11, 15, 21, 24) and (25, 35, 45, 55, 80, 90, 100).

the number of key comparisons needed is

- a) 9
- b) 5
- c) 7
- d) 6
- e) None of the above

Questions 25 to 28 refer to the following Longest Common Subsequence problem.

Let $c[i, j]$ be the length of the Longest Common Subsequence of $X_i = x_1, x_2, \dots, x_i$ and $Y_j = y_1, y_2, \dots, y_j$. Then $c[i, j]$ can be recursively defined as following:

$$c[i, j] = \begin{cases} 0 & \text{if } i = 0 \text{ or } j = 0 \\ c[i-1, j-1] + 1 & \text{if } i, j > 0 \text{ and } x_i = y_j \\ \max\{c[i-1, j], c[i, j-1]\} & \text{if } i, j > 0 \text{ and } x_i \neq y_j \end{cases}$$

The following is an incomplete table for the sequences of AATGTT and AGCT.

		A	A	T	G	T	T
	0	0	0	0	0	0	0
A	0	1	1	1	1	1	1
G	0	1	1	1	2	2	2
C	0	1	1	1	2	2	2
T	0	1	1	2	2	3	3

25. The value of $c[3, 4]$ is

- a) 1
- b) 2
- c) 3
- d) 4
- e) 5

26. The length of the longest common subsequence of AATGTT and AGCT is

a) 1 b) 2 c) 3 d) 4 e) 5	
27. The longest common subsequence of AATGTT and AGCT is	
a) AGCT b) ATGT c) AATG d) AGC e) AGT	
28. The length of the longest common subsequence of AATGT and AGC is	
a) 1 b) 2 c) 3 d) 4 e) 5	

PART B (40 marks)

Question 1 (13 marks)

1. Suppose $T(n)$ denotes the worst case time complexity of the merge-sort algorithm on n numbers.

- a) Explain why $T(n)$ can be described by the following recurrence. 7

$$T(n) = \begin{cases} 1 & \text{if } n = 1 \\ 2T(\lfloor n/2 \rfloor) + n & \text{if } n > 1 \end{cases}$$

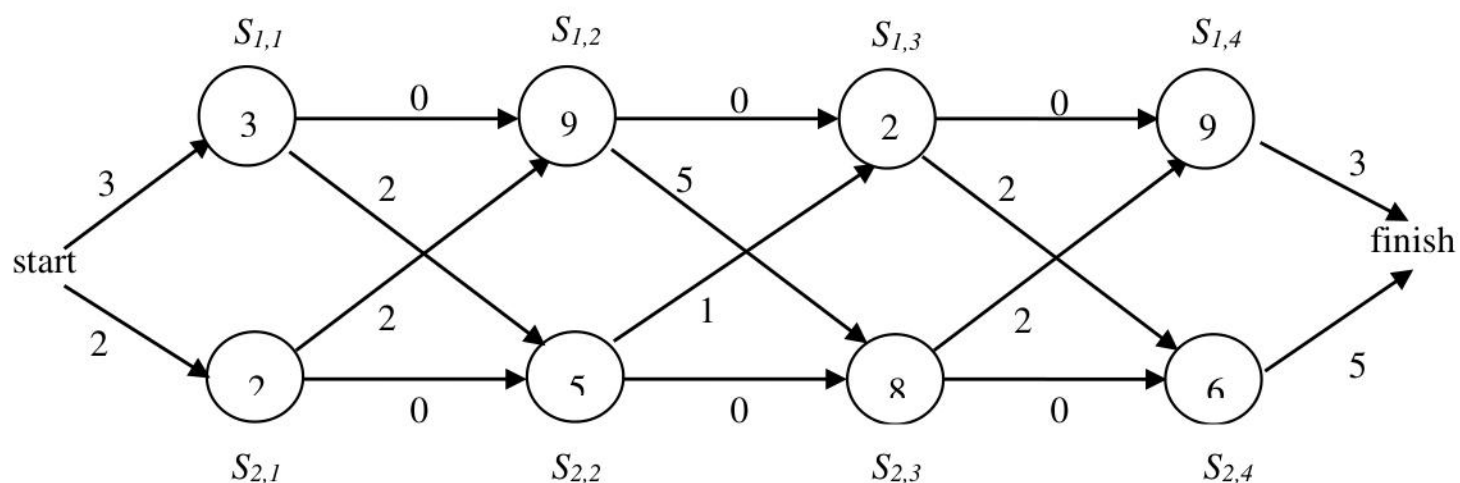
- b) Show that $T(n) = O(n \log n)$ by the substitution method. (Hint: show that $T(n) \leq 2n \log n$ for $n \geq 2$ by Mathematical Induction.) 6

Question 2 (27 marks)

Suppose there are two assembly lines each with n stations, for each j , $1 \leq j \leq n$, let $S_{i,j}$ denote the j th station in line i ($i = 1, 2$). For each i ($i = 1, 2$), let

- e_i be the entry time into line i .
- x_i be the exit time from line i .
- $a_{i,j}$ be the assembly time at $S_{i,j}$
- $t_{1,j}$ be the transfer time from station $S_{1,j}$ to station $S_{2,j+1}$ and $t_{2,j}$ be the transfer time from station $S_{2,j}$ to station $S_{1,j+1}$.

The following is an example of two assembly lines, each line has 4 stations with $e_1=3$, $e_2=2$, $x_1=6$, $x_2=5$. The assembly time is given in the circle representing the station and the transfer time is given next to the edge from one station to another.



1. Let $f_1[j]$ and $f_2[j]$ be the fastest time to get through stations $S_{1,j}$ and $S_{2,j}$, respectively. Using dynamic programming technique, derive a recursive definitions of $f_1[j]$ and $f_2[j]$ and the fastest time f^* needed to get through the assembly line. 6

2. For the assembly lines give above, fill in the table of the fastest time $f_i[j]$ needed to get through station $S_{i,j}$. Show all the intermediate steps in computing these values. **6**

j	$f_1[j]$	$f_2[j]$
1	...	...
2	...	...
3	...	...
4	...	...

3. For the assembly lines give above, what is the fastest time f^* needed to get through the assembly line? **2**
4. For the assembly lines give above, which stations should be chosen? **6**
5. Based on the recurrence relation obtained in a), write a pseudo code of the bottom-up dynamic programming algorithm with running time $O(n)$ for the n stations assembly line scheduling problem. **7**

END OF THE PAPER